Camille Fourmaintraux

Application for a Software Development Internship

Profile

Fourth-year engineering student in Computer Science at Polytech Grenoble (Université Grenoble Alpes, France) and graduate of a *BUT Informatique*. Six months of internship experience as a software developer at E-Mothep, where I was rehired to maintain and extend a monitoring tool used in production by large clients: REST API development, new features, bug fixing, UI redesign and documentation. Full-stack experience with Java, Spring Boot, React and TypeScript, and Scrum Master in Agile teams of up to seven people. Available from around April 2027 for an internship of at least sixteen weeks, as part of my engineering curriculum.

Education

- **2025 – Present**
  - Engineering Degree (*Diplôme d'ingénieur*) in Computer Science, 4th year, Polytech Grenoble, Université Grenoble Alpes
  - Coursework: software engineering, hardware architecture, computer networking, signal processing, numerical methods, game engine development, functional programming (OCaml), formal proof (Rocq)
- **2025:** *BUT Informatique* (Bachelor of Technology in Computer Science), IUT de Lille
- **2024:** One-month English language program, Kaplan International, London
- **2022:** French Baccalauréat (General, with Honors), Lycée Pierre d'Ailly, Compiègne

Projects

1. **2D Game Engine** *(Java, GitLab)*
   - Built a complete 2D game engine in a six-person team (end-of-year project)
   - Built a game with the engine to test and showcase its capabilities
2. **Reinforcement Learning Environment** *(Java)*
   - Designed a custom environment and a basic RL agent to benchmark it
3. **CodeSwipe, Full-Stack Web Application** *(Java, Spring Boot, Spring Security, React)*
   - Tinder-style app where users swipe on code snippets instead of profiles
   - Developed the back-end (REST API, JWT authentication) and the front-end
   - Scrum Master of a seven-person team (sprint tracking, tickets, Git workflow)
4. **Real-Time Multiplayer Game** *(JavaScript, Node.js, Socket.IO, HTML5 Canvas)*
   - Built the whole game as a pair: engine and multiplayer server with game rooms
5. **Appointment Booking Website** *(Spring Boot, Spring Security, JPA, React, Tailwind CSS)*
   - Web application with role-based access control (user and admin), built as a pair
6. **RISC-V Processor Emulator** *(C)*
   - Simulated registers, ALU operations, memory and the instruction execution cycle
7. **Digital Portfolio, Personal Project** *(React, TypeScript, Vite, Node.js)*
   - Bilingual website with a contact form and an AI chatbot about my profile

Work Experience

- **Software Development Intern, E-Mothep (April – June 2024 and March – June 2025)**
  SOA integration consulting firm (webMethods), with clients such as Decathlon and GIFI
  - Rehired for a second internship after a successful first one (26 weeks in total)
  - Maintained and extended AMAT, a production tool monitoring clients' webMethods environments and reporting anomalies
  - Built a REST API letting clients update their monitoring references, with very positive client feedback
  - Added monitoring modules (license expiry warnings, server settings) and automated the creation of monitoring references
  - Redesigned the web interface (dynamic forms, accessibility) and centralized the technical documentation
- **Tutor, University of Lille (October 2024 – February 2025)**
  - Supported first-year students in mathematics, programming and databases (SQL)
- **General Store Employee, Carrefour Contact (July – August 2022)**

Contact

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Skills

Programming Languages
Java, JavaScript, TypeScript, Python, C, Kotlin, OCaml

Back-End
Spring Boot, Spring Security, JPA/Hibernate, Java EE (Servlet, JSP), Node.js, Socket.IO

Front-End
React, Redux, Angular, HTML/CSS, Tailwind CSS, Bootstrap, Vite

Databases
SQL, PostgreSQL, MongoDB, Redis, Firebase, DBeaver

APIs and Integration
REST, SOAP, JWT, webMethods, Software AG Designer, Postman, Bruno

Tools and DevOps
Git, GitHub, GitLab CI/CD, Mantis, Trello, LaTeX

Methodology
Agile Scrum (Scrum Master on two projects)

Languages
French: native
English: advanced (TOEIC C1, 2025)

Other
Driver's license (Category B)

Qualities

Organized, attentive, curious, patient

Interests

Video game programming (Godot, game jams), member of a board game club, hiking, fantasy novels, puzzles and escape rooms

Portfolio

My website details all my academic, professional and personal projects.